

CH # 4 Automata Solutions

Q.2 - R.E = $(aaa + b)^*$

$L = \{aaa, b, bb, bbb, baaa, aaab, bbbb, baaab, \dots\}$

Q.3 - R.E = $(a+b)^* (S_1 + S_2 + S_3 + S_4)^+ (a+b)^*$

$L = \{S_1, S_2, S_3, S_4, a, b, aS_1, bS_1, aS_2, \dots\}$

Q.4 R.E = $a^* b a^* b a^* b a^*$

$L = \{bbb, abbb, babb, bbab, bbbab, ababb, \dots\}$

Q.5 R.E = $a^* b a^* b a^* + a^* b a^* b a^* b a^*$

$L = \{bb, bbb, abb, bab, bba, abbb, bbbab, \dots\}$

Q.6 (i) R.E = $(a+b)^+ (aa+bb)$

$L = \{abb, aqa, qa, bb, ababb, baabb, \dots\}$

(ii) R.E = $[(1+b)(ab)^*] aq [(ba)^*(1+b)] +$

$[(1+a)(ba)^*] bb [(ab)^*(1+a)]$

$L = \{aa, bb, baab, abba, abb, bba, baqa, \dots\}$

Q.7 R.E = $(1+b+bb)(a+ab+abb)^*$

$L = \{1, a, b, ab, ba, aa, bb, abb, bba, aqa, \dots\}$

Q.8 R.E = $(1+b+bb)(a+ab+abb)^* + (1+a+aa)(b+ba+baa)^*$

$L = \{1+a, b, aqa, bb, bbbqa, \dots\}$

Q.9 (i) R.E = $b^* a^*$

$L = \{ \Lambda, a, b, ba, bba, baa, bbaa \dots \}$

Q.9 (ii) $(\Lambda + b)(a + ab)^*$

$L = \{ \Lambda, a, b, ab, ba, aab, baa, abab, aabab \dots \}$

Q.10 R.E = $(b^* a b^* a b^* a b^*)^*$

$L = \{ \Lambda, aab, baab, abab, aabab, baabab \dots \}$

Q.11 (i) $a^* b (bb)^* a^* + (a^* b (bb)^* a a^*)^* + (a a^* b (bb)^* a)^*$
 $L = \{ b, ab, ba, abab, abbbab, ababbbab \dots \}$

(ii)

R.E = $[aa + bb + (ab + ba)(aa + bb)^*(ab + ba)]^* (b + abab)$

$[aa + bb + (ab + ba)(aa + bb)^*(ab + ba)]^*$
 $L = \{ b, aab, baab, abab, abababab \dots \}$

(iii)

R.E = $[(aa + bb)^*(ab + ba)(aa + bb)^*] [ab + ba] (aa + bb)^* (ab + ba)$

$L = \{ ab, aabbbb, ababab, aaaa \dots \}$

[Signature]